



TEAM NAME: Devhouse

COLLEGE NAME: ML-Powered Postpartum Maternal Health

MonitorPROBLEM STATEMENT: New mothers suffer from unnoticed severe issues like anemia, jaundice, and cardiovascular problems due to unintentional shift in care toward newborns, incurring travel expenses while their health remains ignored.

PROBLEM STATEMENT

Problem : New mothers suffer from unnoticed severe issues like anemia, jaundice, and cardiovascular problems due to unintentional shift in care toward newborns, incurring travel expenses while their health remains ignored.



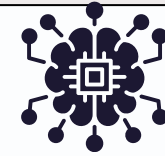
Postpartum Anemia



Untreated postpartum anemia causes **fatigue, weakness, slow recovery** affecting infant care.



Jaundice Risk



Undetected postpartum jaundice causes **liver damage, bleeding complications, maternal emergencies.**



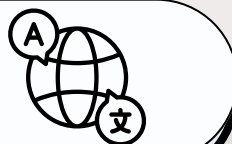
Cardiac Strain



Postpartum cardiovascular stress **increases risk of heart failure, stroke, death.**



Mental Health



Postpartum depression **disrupts emotional stability**, bonding, caregiving, infant development



Limited Awareness



Lack of maternal health awareness prevents early detection and treatment.



Missed Monitoring



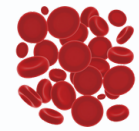
Limited postpartum monitoring leaves **complications unnoticed**, **worsening** long-term maternal health.

SOLUTION

Approach:

We empower mothers through a government-integrated app: at-home screening instantly alerts community health workers, enabling continuous clinical monitoring and truly connected maternal care.

AI Anemia Screening



Computer vision analyzes conjunctival images estimating hemoglobin levels, reducing invasive blood testing.

Jaundice Detection



AI scleral imaging identifies bilirubin increase early, preventing severe liver complications.

Cardiac Monitoring



PPG-based monitoring detects postpartum cardiovascular stress, enabling early medical intervention and treatment.

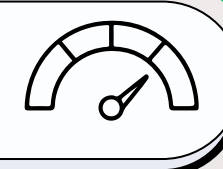
Mental Health Support



Multilingual AI chatbot screens postpartum depression, providing emotional support and guidance.

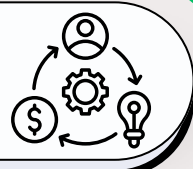


Maternal Awareness



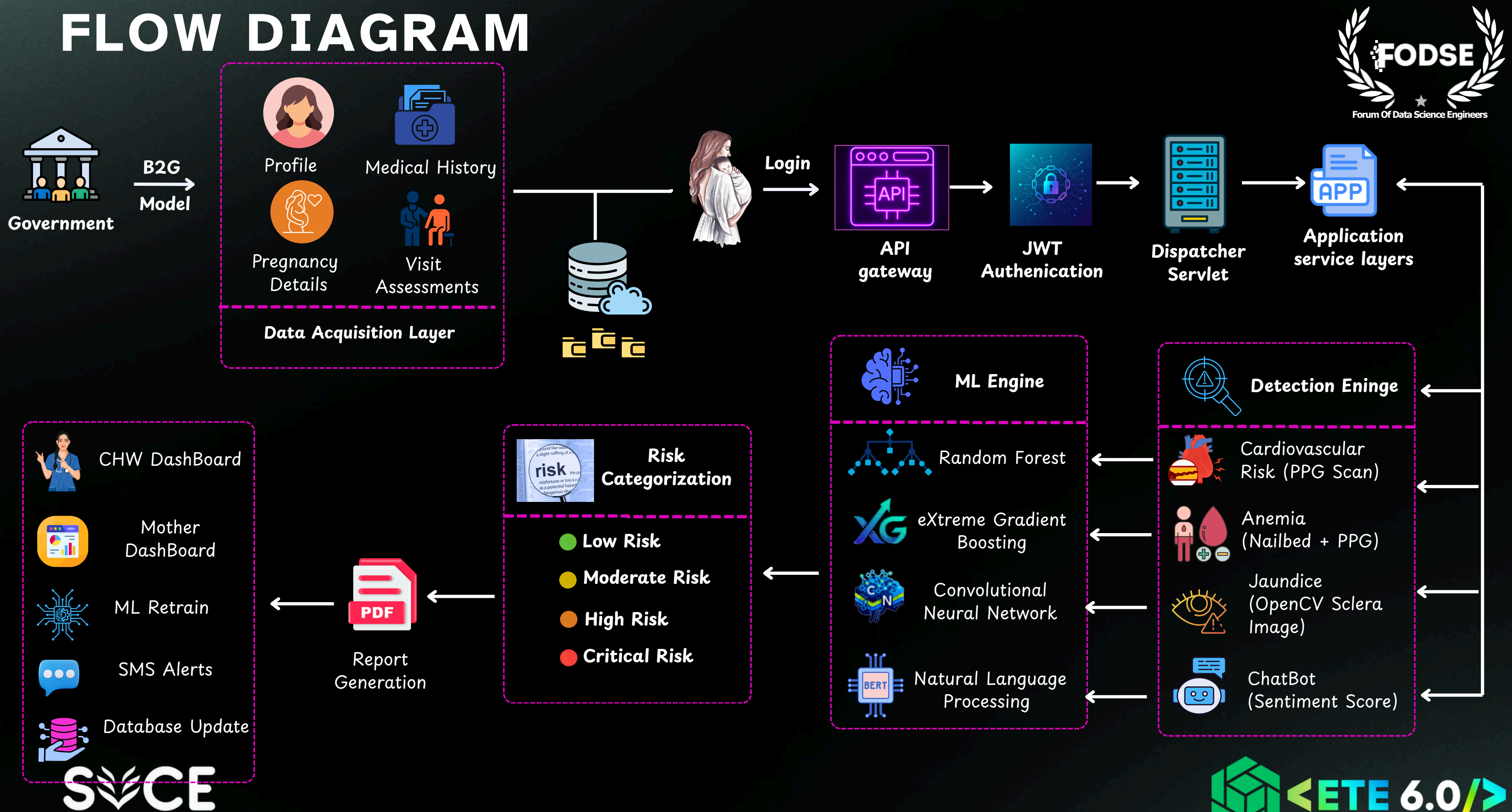
Digital education platform improves maternal awareness enabling early symptom recognition and care.

Offline Storage Alerts

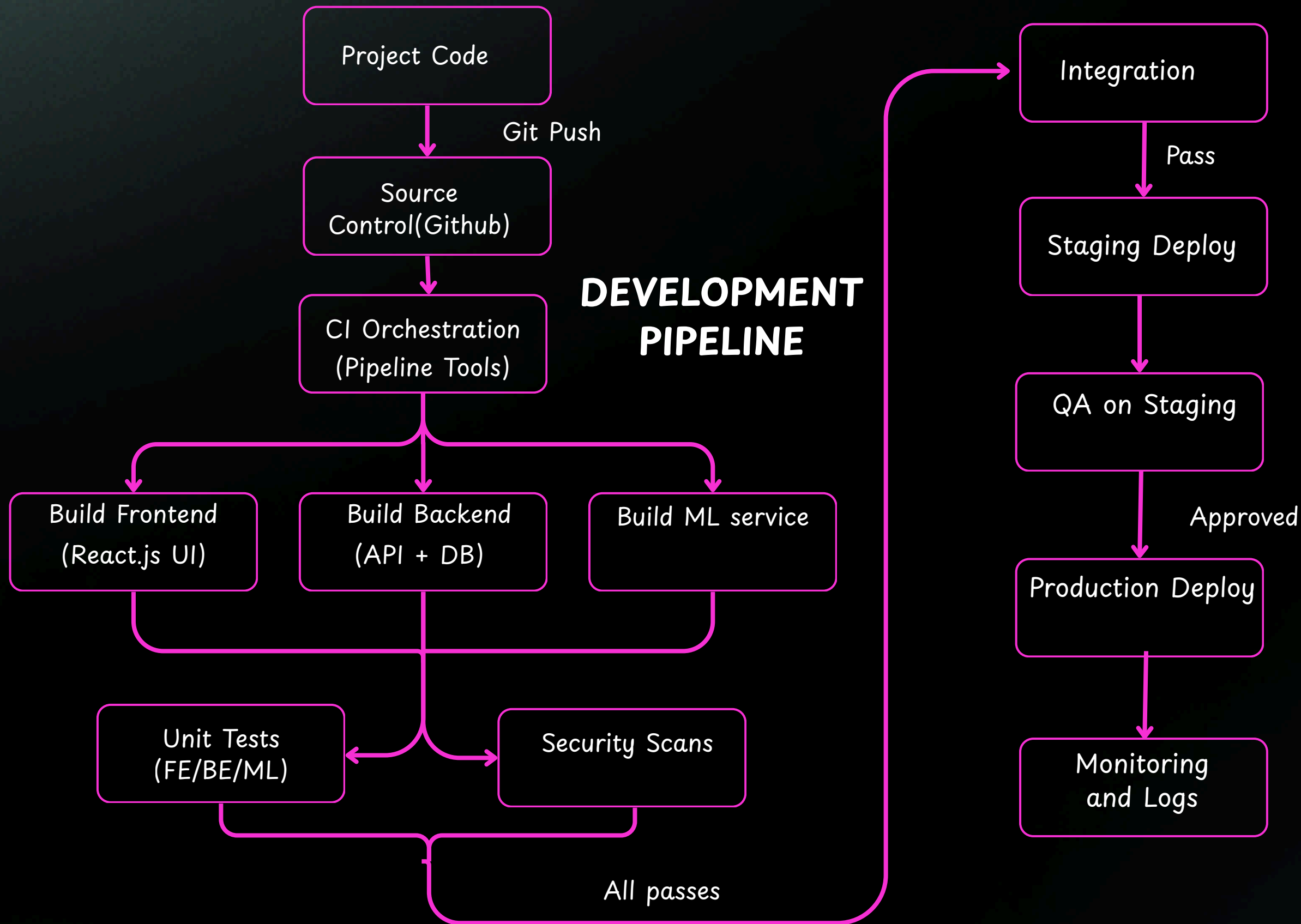


Offline data storage with automated alerts ensures timely health monitoring and follow-ups.

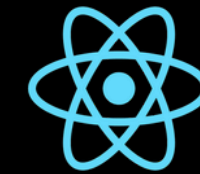
FLOW DIAGRAM



TECHNICAL REQUIREMENTS



TECH STACK



React



Tailwind CSS



capacitor



RBAC



JWT



OAuth

XGB

CNN

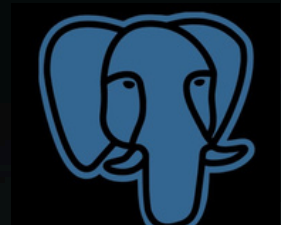


NLP



uvicorn

FastAPI



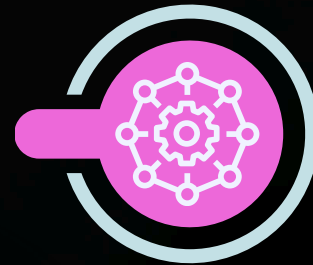
PostgreSQL

NOVELTY / USE CASE

NOVELTY

Zero-Cost Model

Free for mothers via district licenses and public health partnerships



Policy Alignment

Works with PMSMA and JSSK programs for smooth state-level healthcare adoption.



Individualized Care

Personalized guidance adapting to each mother's health journey.



Nationwide Scalability

Deploys via ASHA networks and PHCs for rapid maternal health expansion.



USE-CASE

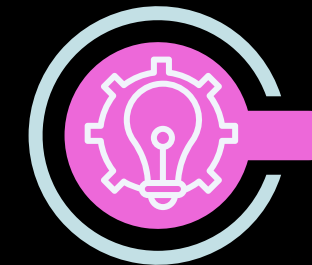
Validated Models

Applies research-proven methods for heart signals, visual checks, health understanding.



Data Security

Protects maternal health information with encryption, keeping personal details safe.



Data Rollback

Restores health records to verified states, ensuring accuracy safely.



Unified Platform

Teleconsultation, mental health, emergency aid, nutrition in one app.



BUSINESS MODEL / TARGET MARKET

Government Entry

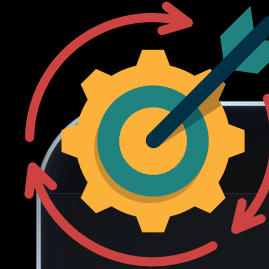


Healthcare workers upload postpartum mothers data through hospital portals immediately after delivery.

Mother Login



Registered mothers access platform using credentials to monitor health status, alerts.



Government Hospitals

Public hospitals adopting digital tools to monitor postpartum mothers and improve maternal outcomes.

Healthcare Workers

Doctors and nurses needing efficient systems to track maternal health risks early.

BUSINESS MODEL

Scheme Integration



Platform supports government schemes like POSHAN Abhiyaan improving maternal healthcare monitoring.

B2G Model



B2G partnership enables institutions deploy platform for nationwide maternal monitoring.

Rural Mother



Postpartum mothers in rural areas requiring accessible digital maternal health monitoring tools.

Health Authorities

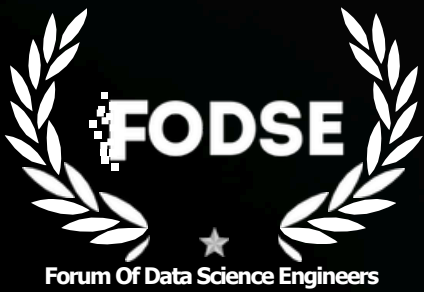


Government health departments responsible for improving maternal healthcare monitoring and policy planning

TARGET MARKET



TEAM DETAILS



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